

Class 10 NCERT Maths

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Chapter 5: Arithmetic Progressions (Core-Basics Edition)

CHAPTER 5: ARITHMETIC PROGRESSIONS

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) SEQUENCE KYA HAI?

Sequence = numbers ki list ek pattern mein.

Jaise 2, 4, 6, 8, ... ya 1, 4, 9, 16, ...

2) TERM

Sequence ka har number ek "term" hai.

Pehla term, doosra term, ... nth term.

3) DIFFERENCE NIKAALNA

Do terms ka antar = baad wala - pehle wala.

5, 9 -> $9 - 5 = 4$. (Negative bhi ho sakta hai.)

4) VARIABLE / FORMULA SAMAJHNA

a = pehla term, d = common difference,

n = term number, a_n = nth term, S_n = n terms ka sum.

5) SUBSTITUTION

Formula mein values daal ke calculate karna.

$a_n = a + (n-1)d$ mein a,d,n daal do.

6) BASIC ALGEBRA (do equations)

Kabhi 2 equations milti hain (a aur d ke liye).

Subtract karke ek variable gayab karte hain.

7) NEGATIVE NUMBERS

d negative ho toh sequence ghatti hai:

10, 7, 4, 1, ... (d = -3).

NOW: ASLI CHAPTER 5 SHURU

TOPIC 1: AP KYA HAI?

AP = sequence jisme har term, pichle term mein ek FIXED number (common difference d) add karke milta hai.

Examples: 2,5,8,11 (d=3) ; 10,7,4 (d=-3) ; 5,5,5 (d=0).

[CORE] d = koi bhi do consecutive terms ka antar (baad wala - pehle wala). Saare antar same hone chahiye.

General form: a, a+d, a+2d, a+3d, ...

TOPIC 2: nth TERM FORMULA

EXAM ALERT: Ye formula har AP question mein use hota:

$$a_n = a + (n - 1) d$$

a = pehla term, d = common diff, n = kaunsa term.

TOPIC 3: SUM OF FIRST n TERMS

EXAM ALERT: Dono sum formulas yaad rakh:

$$S_n = (n/2) [2a + (n-1)d] \quad (a \text{ aur } d \text{ se})$$

$$S_n = (n/2) (a + l) \quad (l = \text{last term})$$

(Doosra formula tab use jab last term l pata ho.)

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples पूरी तरह solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Sum diya, n nikaalo

(Exam favourite - quadratic ban jaata hai)

AP 2, 5, 8, ... ke kitne terms ka sum 155 hai?

$a=2, d=3, S_n=155$.

$$155 = \frac{n}{2}[2(2) + (n-1)3]$$

$$310 = n[4 + 3n - 3] = n[3n + 1]$$

$$3n^2 + n - 310 = 0$$

Split ($ac = -930$, sum = 1 $\rightarrow$ 31, -30):

$$3n^2 + 31n - 30n - 310 = 0$$

$$(3n + 31)(n - 10) = 0$$

$$n = 10 \text{ (positive lo)}$$

Answer: 10 terms.

Solved Example 2 - Do terms diye, AP nikaalo

(Exam favourite - 2 equations)

AP ka 4th term 0 hai aur 11th term -7 hai. AP nikaalo.

$$a + 3d = 0 \quad \dots(1)$$

$$a + 10d = -7 \quad \dots(2)$$

$$(2)-(1): 7d = -7 \rightarrow d = -1$$

$$(1): a = -3d = 3$$

AP: 3, 2, 1, 0, -1, ...

Solved Example 3 - Kaunsa term diya value pe

100, 95, 90, ... ka kaunsa term -50 hai?

$a=100, d=-5, a_n=-50$.

$$-50 = 100 + (n-1)(-5)$$

$$-150 = -5(n-1) \rightarrow n-1 = 30 \rightarrow n = 31$$

Answer: 31st term.

Solved Example 4 - Sum of n terms

2, 5, 8, ... ke first 20 terms ka sum.

$a=2, d=3, n=20.$

$$S_{20} = (20/2)[2(2) + 19(3)] = 10[4 + 57] = 610.$$

Solved Example 5 - Word problem (seats/salary)

(Exam favourite)

Theatre ki first row mein 20 seats, har agli row mein 2 zyada. 30th row mein kitni seats?

$a=20, d=2, n=30.$

$$a_{30} = 20 + 29(2) = 20 + 58 = 78 \text{ seats.}$$

Solved Example 6 - nth term basic

7, 13, 19, 25, ... ka 50th term.

$a=7, d=6.$

$$a_{50} = 7 + 49(6) = 7 + 294 = 301.$$

Solved Example 7 - Sum 1 to 100 (Gauss)

1+2+3+...+100 ka sum.

$a=1, l=100, n=100.$

$$S_{100} = (100/2)(1 + 100) = 50 \times 101 = 5050.$$

Solved Example 8 (EASIEST) - d nikaalo, AP check

Kya 3, 7, 11, 15 AP hai? d kya?

$7-3=4, 11-7=4, 15-11=4.$ Saare antar same.

Haan AP hai, $d = 4.$

Q AND A TIME

Q1. CORE: kya 1, 4, 9, 16 AP hai? Kyun/kyun nahi?

Q2. AP 5, 9, 13, ... ka 20th term nikaal.

Q3. AP ka 10th term 32, 16th term 56 hai. d nikaal.

Q4. AP 3, 7, 11, ... ke first 15 terms ka sum.

Q5. 10, 7, 4, ... ka kaunsa term -20 hai?

Q6. AP 2, 5, 8, ... ke kitne terms ka sum 95 hai?

Q7. CORE CHECK: $a=4$, $d=3$ wali AP ke pehle 4 terms likho.

SUMMARY

1. AP: har term mein fixed d add hota hai.
2. n th term: $a_n = a + (n-1)d$.
3. Sum: $S_n = \frac{n}{2}[2a + (n-1)d]$ ya $\frac{n}{2}(a + l)$.
4. d = baad wala term - pehle wala (sab same hone chahiye).
5. Word problems: a , d , n pehchaano $\rightarrow$ formula lagao.

CORE (RED page) revise: sequence, term, d nikaalna, substitution, negative numbers.

Generated by Kiro for Krishna1k

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